RGB ELEKTRONIKA
Centrum automatyki przemysłowej



DATASHEET

SEMIKRON

SKIIP 82AC12

OTHER SYMBOLS:

SKIIP82AC12, SKIIP 82AC12

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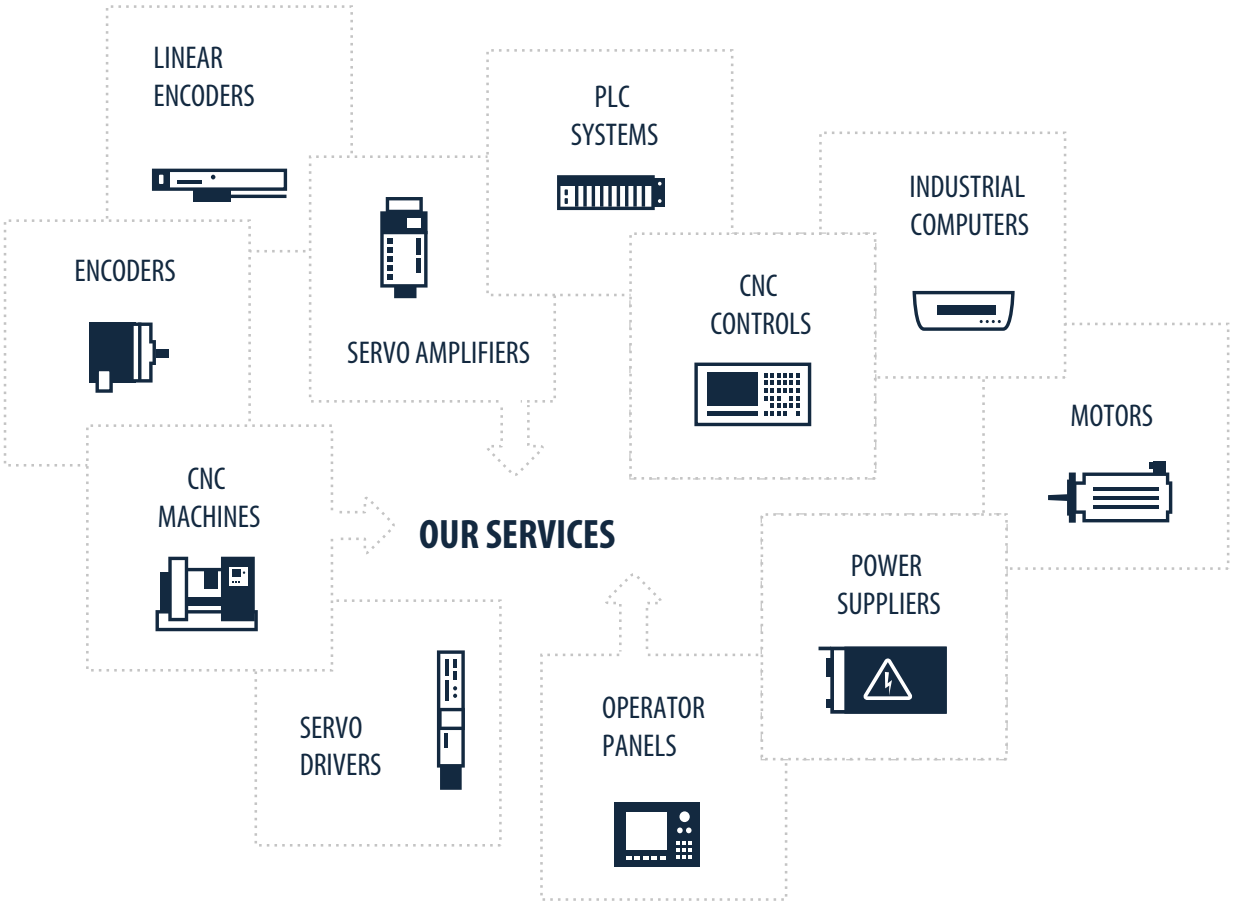
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SKiiP 82 AC 12 - SKiiP 82 AC 12 I

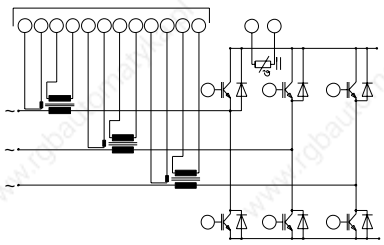
Absolute Maximum Ratings Symbol	Conditions [1]	Values	Units
V_{CES}		1200	V
V_{GES}		± 20	V
I_C	$T_{heatsink}$ = 25 / 80 °C	95 / 65	A
I_{CM}	t_p < 1 ms; $T_{heatsink}$ = 25 / 80 °C	190 / 130	A
T_j		– 40 . . . + 150	°C
T_{stg}		– 40 . . . + 125	°C
V_{isol}	AC, 1 min.	2500	V
Inverse Diode			
$I_F = -I_C$	$T_{heatsink}$ = 25 / 80 °C	80 / 53	A
$I_{FM} = -I_{CM}$	t_p < 1 ms; $T_{heatsink}$ = 25 / 80 °C	160 / 106	A
I_{FSM}	t_p = 10 ms; sin., T_j = 25 °C	720	A
I^2t	t_p = 10 ms; sin., T_j = 25 °C	2600	A^2s

Characteristics Symbol	Conditions [1]	min.	typ.	max.	Units
IGBT - Inverter					
V_{CEsat}	I_C = 75 A T_j = 25 (125) °C	–	2,5(3,1)	3,0(3,7)	V
$t_{d(on)}$	V_{CC} = 600 V; V_{GE} = ± 15 V	–	35	70	ns
t_r	I_C = 75 A; T_j = 125 °C	–	70	140	ns
$t_{d(off)}$	$R_{gon} = R_{goff}$ = 15 Ω	–	450	600	ns
t_f	inductive load	–	70	100	ns
$E_{on} + E_{off}$		–	18	–	mJ
C_{ies}	V_{CE} = 25 V; V_{GE} = 0 V, 1 MHz	–	5,0	–	nF
R_{thjh}	per IGBT	–	–	0,35	K/W
Diode [2] - Inverter					
$V_F = V_{EC}$	I_F = 75 A T_j = 25 (125) °C	–	2,0(1,8)	2,5(2.3)	V
V_{TO}	T_j = 125 °C	–	1,0	1,2	V
r_T	T_j = 125 °C	–	11	15	mΩ
I_{RRM}	I_F = 75 A, V_R = – 600 V	–	45	–	A
Q_{rr}	di_F/dt = – 800 A/µs	–	11	–	µC
E_{off}	V_{GE} = 0 V, T_j = 125 °C	–	3,0	–	mJ
R_{thjh}	per diode	–	–	0,8	K/W
Current sensor for three phase output ac current (SKiiP 82 AC 12 I)					
$I_{p\ RMS}$	Continuous current, T = 100 °C, V_{suppl} = ± 15 V	–	50	–	A
$I_{pmax\ RMS}$	t ≤ 2 s	–	–	80	A
$I_{p\ peak}$	t ≤ 10 µs	–	1000	–	A
R_{out}	terminating resistance	–	50	–	Ω
$I_{s\ RMS}$	rated sensor current at I_p = 50 A_{RMS}		25		mA
$I_p : I_s$	transfer ratio		1 : 2000		
Offset $_{error}$	I_P = 0 A, T = – 40 ... 100 °C	–	± 0,2	–	mA
Linearity		–	0,1	–	%
delay time	I_P = 10 % – 80 %	–	< 1	–	µs
	90 % – 20 %	–	< 1	–	µs
Bandwidth			0 – 100	(– 3dB)	kHz
Temperature Sensor					
R_{TS}	T = 25 / 100 °C		1000 / 1670		Ω
Mechanical Data					
M_1	case to heatsink, SI Units	2,5	–	3,5	Nm
Case	mechanical outline see pages B 16 – 11 and B 16 – 12		M8		

MiniSKiiP 8
SEMIKRON integrated intelligent Power
SKiiP 82 AC 12
SKiiP 82 AC 12 I [3]
IGBT
3-phase bridge inverter

Case M8





UL recognized file no. E63532

- more detailed characteristics of current sensors and temperature sensor please refer to part A
- common characteristics see page B 16 – 4

[1] $T_{heatsink}$ = 25 °C, unless otherwise specified
[2] CAL = Controlled Axial Lifetime Technology (soft and fast recovery)
[3] With integrated closed loop current sensors

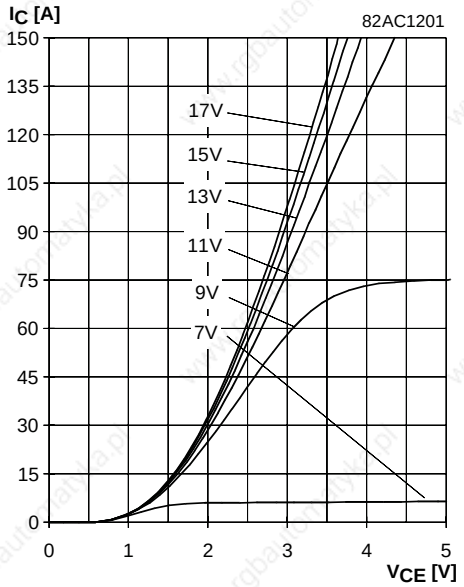


Fig. 1 Typ. output characteristic, t_p = 80 µs; 25 °C

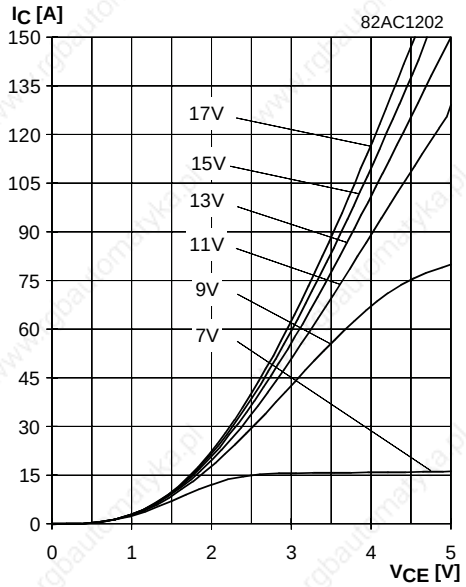
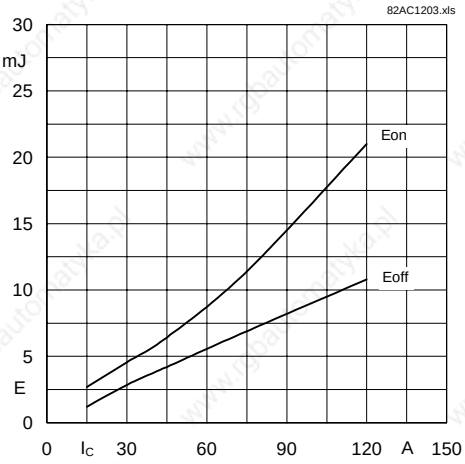
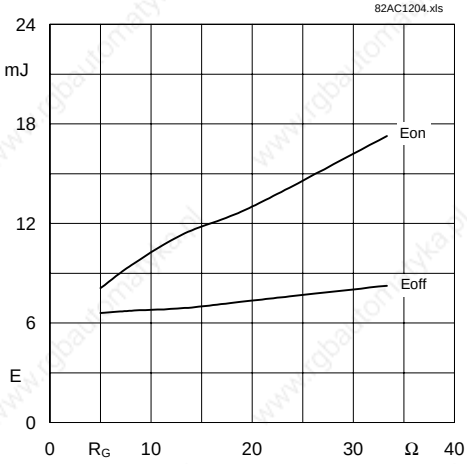


Fig. 2 Typ. output characteristic, t_p = 80 µs; 125 °C



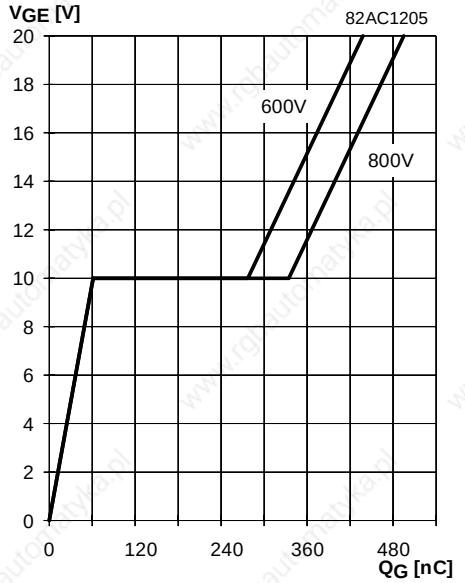
T_j = 125 °C
V_{CE} = 600 V
V_{GE} = ± 15 V
R_G = 15 Ω

Fig. 3 Turn-on /-off energy = f (I_C)



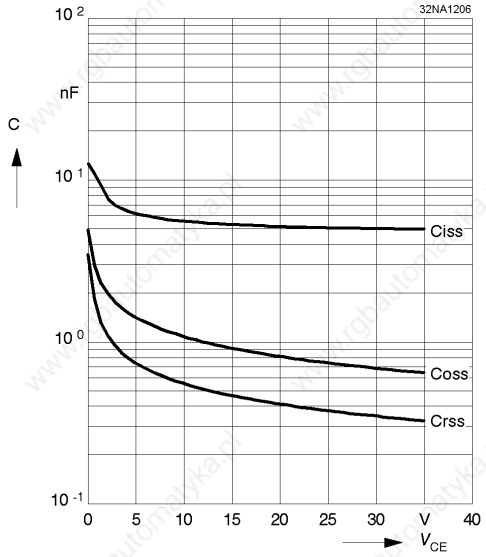
T_j = 125 °C
V_{CE} = 600 V
V_{GE} = ± 15 V
I_C = 75 A

Fig. 4 Turn-on /-off energy = f (R_G)



I_{Cpuls} = 75 A

Fig. 5 Typ. gate charge characteristic



V_{GE} = 0 V
f = 1 MHz

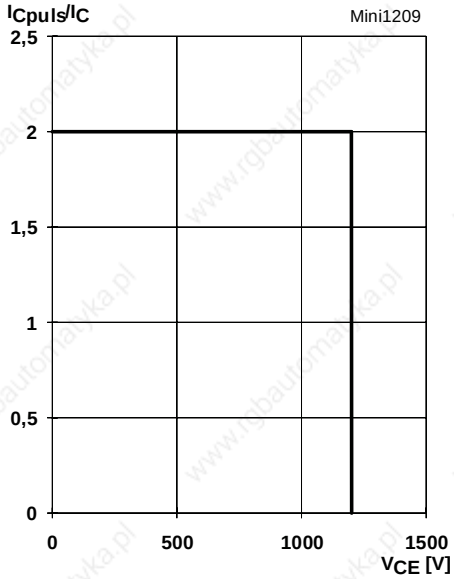
Fig. 6 Typ. capacitances vs. V_{CE}

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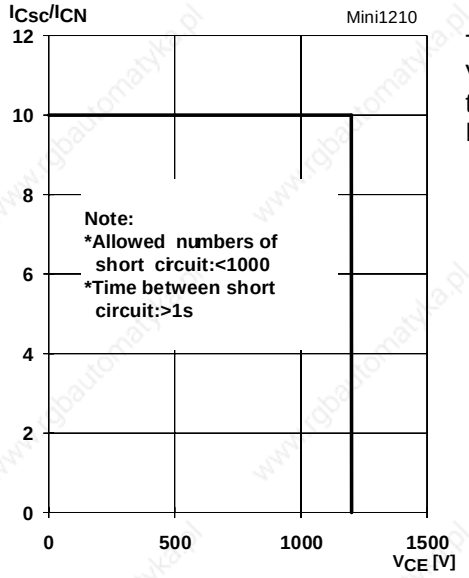
T_j = 150 °C
V_{GE} = ≥ 15 V

Fig. 7 Rated current of the IGBT $I_{Cop} / I_C = f(T_h)$



T_j = ≤ 150 °C
V_{GE} = ± 15 V

Fig. 9 Turn-off safe operating area (RBSOA) of the IGBT



T_j = ≤ 150 °C
V_{GE} = ± 15 V
t_{sc} = ≤ 10 µs
L_{ext} < 25 nH

Fig. 10 Safe operating area at short circuit of the IGBT

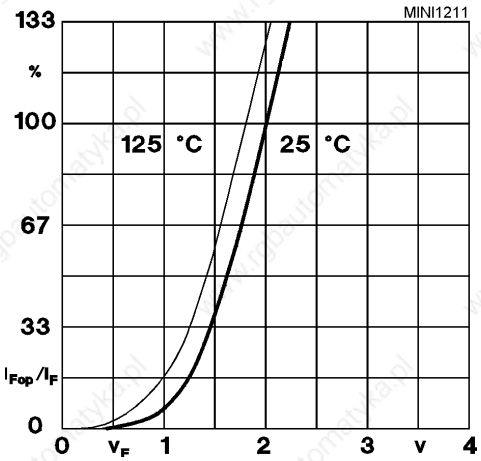


Fig. 11 Typ. freewheeling diode forward characteristic

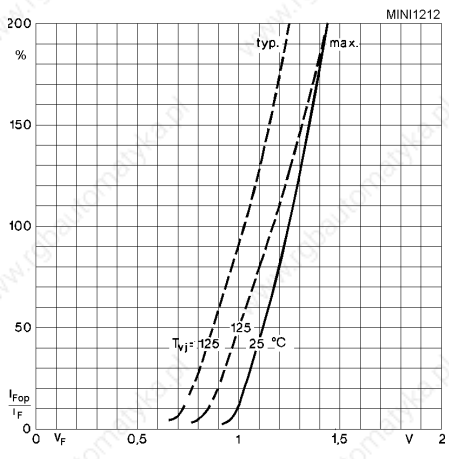


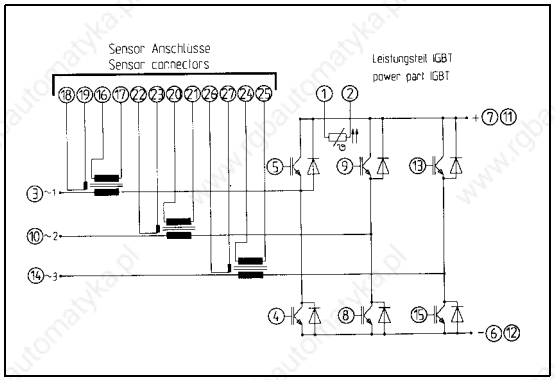
Fig. 12 Forward characteristic of the input bridge diode

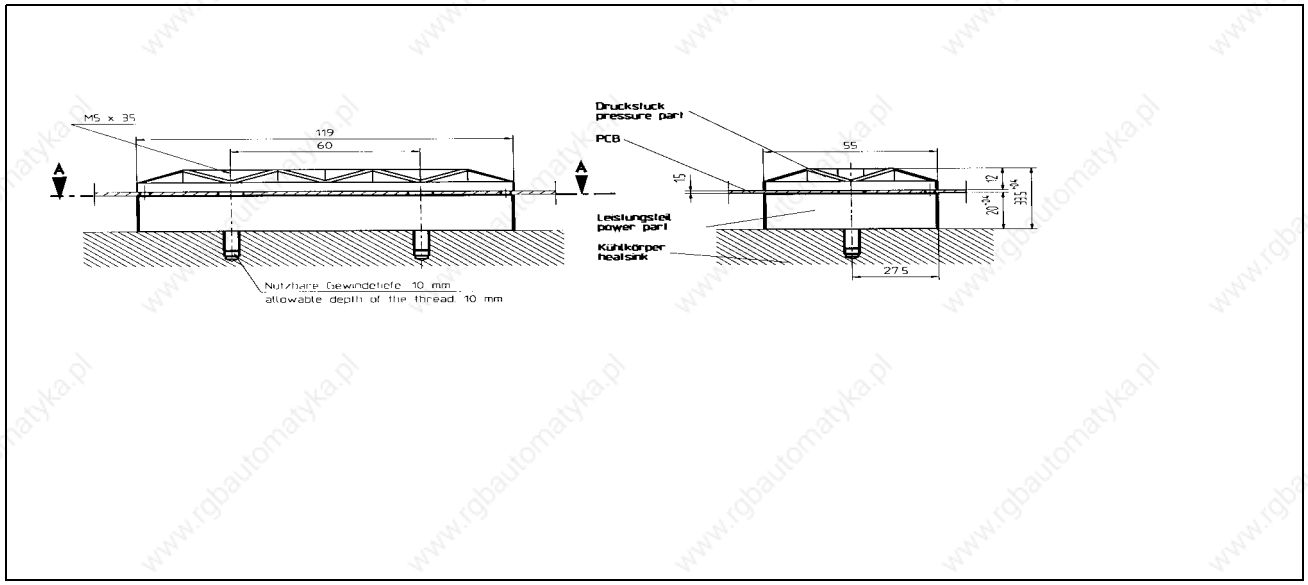
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MiniSKiiP 8

Inverter part

SKiiP 82 AC 06 ...	Circuit
SKiiP 83 AC 06 ...	Case M8
SKiiP 81 AC 12 ...	Note: The current sensors are
SKiiP 82 AC 12 ...	available only by option I
SKiiP 83 AC 12 ...	



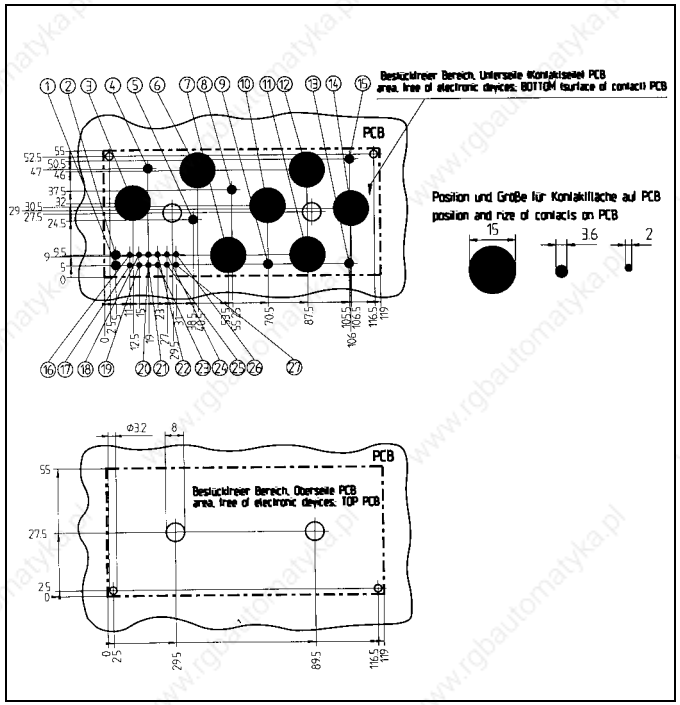


MiniSKiiP 8

Inverter part

SKiiP 82 AC 06 ...
SKiiP 83 AC 06 ...
SKiiP 81 AC 12 ...
SKiiP 82 AC 12 ...
SKiiP 83 AC 12 ...

Case M8
Layout and connections for the customer's printed circuit board



pin		connection	optional
1		T +	
2		T -	
3	~1	ET1 CB1	
4		GB1	
5		GT1	
6	–	EB1 EB2 EB3	
7	+	CT1 CT2 CT3	
8		GB2	
9		GT2	
10	~2	ET2 CB2	
11	+	CT1 CT2 CT3	
12	–	EB1 EB2 EB3	
13		GT3	
14	~3	ET3 CB3	
15		GB3	
16		K1 for ~1	X
17		K2 for ~1	X
18		S1 for ~1	X
19		S2 for ~1	X
20		K1 for ~2	X
21		K2 for ~2	X
22		S1 for ~2	X
23		S2 for ~2	X
24		K1 for ~3	X
25		K2 for ~3	X
26		S1 for ~3	X
27		S2 for ~3	X